



System Architecture & Engineering Blueprint

C4 Diagrams, State Machine & Advisory Locking Sequences

ARCHITECTURE BLUEPRINT

Effective Date: September 16, 2026
System: Order Control System v1.0

System Architecture & Engineering Blueprint

Order Control System — Multi-Brand Dark Kitchen Operating Platform

Printable Edition: [Download Executive PDF \(A4 Edition\)](#)
Architecture Approval Date: September 16, 2026 | Status: Production-Grade & Fully Implemented

Table of Contents

- 1. [Core Architectural Invariants](#)
- 2. [C4 Model — Level 1: System Context](#)
- 3. [C4 Model — Level 2: Containers & Components](#)
- 4. [Order State Machine Architecture](#)
- 5. [Rush Hour Concurrency & PostgreSQL Advisory Locking](#)
- 6. [Zero-Drift Financial Accounting Flow](#)
- 7. [Role-Based Access Control & Audit Invariants](#)

1. Core Architectural Invariants

The Order Control System is the centralized operational platform for a multi-brand sushi dark kitchen operating 4 virtual brands (Flower Sushi, Mastery Sushi, Niwa Sushi, Tobiko Sushi) across 6 aggregator channels. The system operates on a 100% delivery model with no dine-in or customer pickup counters. The architecture is governed by five mandatory engineering directives:

- 1. **Zero Business Logic in UI:** All financial calculations, state machine transitions, and validation rules reside exclusively in `src/lib/`. Database mutations are strictly isolated in `src/services/`. UI components are pure presentation layers that dispatch actions and render state.
- 2. **Single Source of Truth:** Direct Prisma client invocations from UI pages or API route handlers are prohibited. Every order mutation must pass through `src/services/orders.ts`, which atomically records an immutable `audit()` log entry within the same interactive database transaction.
- 3. **Strict Immutability & No Hard Deletes:** No business record (Order, Customer, AuditLog) can ever be deleted. Cancellation is the only terminal state for orders and strictly requires an explicit `CancelReason`. Catalog entities enforce soft-deletion via `isActive = false`.
- 4. **Zero-Drift Financial Identity:** The financial equation `totalCash + totalVisa + totalOnline == totalRevenue` is mathematically enforced and fuzz-tested across 10,000+ randomized orders.

5. **Concurrency-Safe Sequencing:** Order numbers (`ORD-YYYYMMDD-XXXX`) are serialized using PostgreSQL transaction-level advisory locks (`pg_advisory_xact_lock`), completely preventing race conditions and sequence collisions during rush hours.

2. C4 Model — Level 1: System Context

The System Context diagram illustrates the system boundaries, human actors, and external aggregator platforms:



Syntax error in text
mermaid version 10.9.8

3. C4 Model — Level 2: Containers & Components

This diagram details the container architecture, communication protocols, and the internal separation of concerns:



Syntax error in text
mermaid version 10.9.8

4. Order State Machine Architecture

The order lifecycle adheres to a deterministic, unidirectional state machine. Backward status transitions are mathematically disallowed:



Syntax error in text
mermaid version 10.9.8

5. Rush Hour Concurrency & PostgreSQL Advisory Locking

To guarantee that multiple cashiers entering orders simultaneously during rush hours never collide or produce duplicate order numbers, the system utilizes PostgreSQL transaction-scoped advisory locks:



Syntax error in text
mermaid version 10.9.8

6. Zero-Drift Financial Accounting Flow

This flowchart illustrates the end-to-end financial calculation pipeline, from item pricing up to the cash drawer closing reconciliation:



Syntax error in text

mermaid version 10.9.8

7. Role-Based Access Control & Audit Invariants

Access Control Matrix

Resource & Operation	Cashier (CASHIER)	Manager (MANAGER)	Owner (OWNER)	Security Enforcement Mechanism
Create & Search Orders	✔ Allowed	✔ Allowed	✔ Allowed	Session authentication
Advance Order Statuses	✔ Allowed	✔ Allowed	✔ Allowed	State Machine validation
Request Order Discount	✔ Allowed (with reason)	✔ Allowed	✔ Allowed	DISCOUNT_REQUEST event
Approve / Reject Discounts	✘ Forbidden	✔ Allowed	✔ Allowed	403 Forbidden
Open Operational Shift	✔ Allowed	✔ Allowed	✔ Allowed	Session authentication
Record Daily Expense	✔ Allowed	✔ Allowed	✔ Allowed	Session authentication
Close Shift & Reconcile Cash	✘ Forbidden	✔ Allowed	✔ Allowed	403 Forbidden
Manage Menu Catalog & Prices	✘ Forbidden	✔ Allowed	✔ Allowed	403 Forbidden
Analytics Dashboards & Reports	✘ Forbidden	✘ Forbidden	✔ Allowed	403 Forbidden
Audit Trail Inspector & Diff Modal	✘ Forbidden	✘ Forbidden	✔ Allowed	403 Forbidden
Staff User Provisioning & Roles	✘ Forbidden	✘ Forbidden	✔ Allowed	403 Forbidden
Hard Delete Any Business Record	⊘ Prohibited	⊘ Prohibited	⊘ Prohibited	405 Method Not Allowed

Official Architecture Specification for Order Control System. Maintained by Core Engineering.

Order Control System — Multi-Brand Dark Kitchen Operating Platform

Confidential & Proprietary © 2026